



10602 - (HeI)ghly Peculiar? Constraining the properties of the Dense Gas Medium for a $z=5.3$ Little Red Dot using MIRI Medium Resolution Spectroscopy

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI-MRS				
	2	ceers_13318_1	MIRI Medium Resolution Spectroscopy	(1) CEERS-13318
	4	ceers_13318_1_bkg	MIRI Medium Resolution Spectroscopy	(2) CEERS-13318-background
	3	ceers_13318_2	MIRI Medium Resolution Spectroscopy	(1) CEERS-13318
	5	ceers_13318_2_bkg	MIRI Medium Resolution Spectroscopy	(2) CEERS-13318-background

ABSTRACT

We propose a 10.1-hr (on source) pilot study with MIRI/MRS spectroscopy to make a comprehensive analysis on the properties of the dense gaseous medium in CEERS-13318, a spectroscopically confirmed Little Red Dot (LRD) with deep absorption on the broad Balmer lines. With the MIRI/MRS

data we will measure the HeI 10830, Paschen-alpha, Paschen-beta, and Paschen-gamma-emission lines with high SNR. Combined with existing data we will (Goal I) place constraints on the gas density, n_H , in the dense gas medium, (Goal II) probe the hydrogen scattering mechanisms in the broad line region (BLR) based on the profiles of the Balmer broad-emission lines compared to the Pa-lines, and (Goal III) make accurate measurements of the dust-attenuation from the Paschen-to-Balmer lines. The aim of this proposal is to establish a methodology for constraining the properties of these dense gas envelopes that appear to be an important feature of LRDs with broad-line absorption, and highlight the potential for MIRI/MRS to further investigate these gaseous environments in high- z LRDs.

OBSERVING DESCRIPTION

We propose to use the JWST MIRI Medium Resolution Spectroscopy (MRS) mode to observe broad Pa-alpha, Pa-beta, Pa-gamma, and He I 10830 in CEERS-13318, a spectroscopically confirmed Little Red Dot. To achieve our science goals, we require sufficient resolution and S/N to decompose the broad emissions with 2 different MRS observations. For our observations we require F560W for target acquisition as this source is brightest in NIRCAM F444W, and we will require 10 groups in FASTGRPVG64 to achieve the requisite S/N for an exposure time of 1776.026 sec. Our observation strategies are as follows:

For the first MRS observation of our target we will require MRS SHORT and LONG in the Short(A) sub-band with 38 groups, 6 integrations, and 1 four-point negative dither orientation optimized for a point source in SLOW1 readout to minimize data volume for an exposure time of 22,265.405 sec. Following the recommended strategies for MRS, we will use simultaneous MIRI F770W imaging to improve astrometric accuracy in FAST1 readout with 50 groups, 3 integrations, and 1 four-point dither for an exposure of 1687.224 sec. Our dedicated background observations will have the same strategy, but with 1 two-point dither for an exposure time of 11,132.703 sec for MRS and 843.612 sec for MIRI F770W.

For the second MRS observation of our target we will require MRS SHORT and LONG in the Long(C) sub-band with 25 groups, 1 integration, and 1 four-point negative dither orientation optimized for a point source in SLOW1 readout to minimize data volume for an exposure time of 2388.992 sec. Following the recommended strategies for MRS, we will use simultaneous MIRI F1130W imaging to improve astrometric accuracy and fill a photometric gap in FAST1 readout with 50 groups, 4 integrations, and 1 four-point dither for an exposure of 2253.332 sec. Our dedicated background observations will have the same strategy, but with 1 two-point dither for an exposure time of 1194.496 sec for MRS and 1126.666 sec for MIRI F1130W.

Proposal 10602 - Targets - (He)lghly Peculiar? Constraining the properties of the Dense Gas Medium for a z=5.3 Little Red Dot using ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CEERS-13318	RA: 14 19 10.8881 (214.7953671d) Dec: +52 47 19.85 (52.78885d) Equinox: J2000		
	Comments: Category= <i>Galaxy</i> Description= <i>[Active galactic nuclei]</i> Extended= <i>NO</i>				
Fixed Targets	(2)	CEERS-13318-background	RA: 14 19 10.8881 (214.7953671d) Dec: +52 47 19.85 (52.78885d) Equinox: J2000		
	Comments: Category= <i>Calibration</i> Description= <i>[Target acquisition test]</i>				

Proposal 10602 - Observation 2 - (HeI)ghly Peculiar? Constraining the properties of the Dense Gas Medium for a z=5.3 Little Red Dot ...

Observation	Proposal 10602, Observation 2: ceers_13318_1													Fri Mar 13 19:10:27 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[ceers_13318_1_bkg (Obs 4)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates		Targ. Coord. Corrections				Miscellaneous				
	(1)	CEERS-13318		RA: 14 19 10.8881 (214.7953671d)										
				Dec: +52 47 19.85 (52.78885d)										
				Equinox: J2000										
		Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO												
Acquisition	#	Target		Filter	Readout Pattern		Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time		Optional ETC ID	
	1	SAME		F560W	FASTGRPAVG64		10	1		1	1776.026		265120	
Template	Primary Channel			Simultaneous Imaging				Imager Subarray			Grating Wheel Direction			
	All MRS			YES				FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	FASTR1	50	3	1	Dither 1	4	12	1687.224		
	1	SHORT(A)	MRSLONG		SLOWR1	38	6	1	Dither 1	4	24	22265.405		
	1	SHORT(A)	MRSSHORT		SLOWR1	38	6	1	Dither 1	4	24	22265.405		

Special Requirements	Group Observations 2, 4, Non-interruptible
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Proposal 10602 - Observation 4 - (HeI)ghly Peculiar? Constraining the properties of the Dense Gas Medium for a z=5.3 Little Red Dot ...

Observation	Proposal 10602, Observation 4: ceers_13318_1_bkg												Fri Mar 13 19:10:27 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [ceers_13318_1 (Obs 2)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	CEERS-13318-background	RA: 14 19 10.8881 (214.7953671d) Dec: +52 47 19.85 (52.78885d) Equinox: J2000										
	Comments: Category=Calibration Description=[Target acquisition test]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					POINT SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F770W	FASTR1	50	3	1	Dither 1	2	6	843.612	
	1	SHORT(A)	MRSLONG		SLOWR1	38	6	1	Dither 1	2	12	11132.703	
	1	SHORT(A)	MRSSHORT		SLOWR1	38	6	1	Dither 1	2	12	11132.703	

Special Requirements	Group Observations 2, 4, Non-interruptible
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Proposal 10602 - Observation 3 - (HeI)ghly Peculiar? Constraining the properties of the Dense Gas Medium for a z=5.3 Little Red Dot ...

Observation	Proposal 10602, Observation 3: ceers_13318_2												Fri Mar 13 19:10:27 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[ceers_13318_2_bkg (Obs 5)]												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	CEERS-13318	RA: 14 19 10.8881 (214.7953671d) Dec: +52 47 19.85 (52.78885d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID			
	1	SAME	F560W	FASTGRPAVG64	10	1		1	1776.026	265120			
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	50	4	1	Dither 1	4	16	2253.332	
	1	LONG(C)	MRSLONG		SLOWR1	25	1	1	Dither 1	4	4	2388.992	
	1	LONG(C)	MRSSHORT		SLOWR1	25	1	1	Dither 1	4	4	2388.992	

Special Requirements	Group Observations 3, 5, Non-interruptible
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Proposal 10602 - Observation 5 - (He)lghly Peculiar? Constraining the properties of the Dense Gas Medium for a z=5.3 Little Red Dot ...

Observation	Proposal 10602, Observation 5: ceers_13318_2_bkg												Fri Mar 13 19:10:27 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [ceers_13318_2 (Obs 3)]												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	CEERS-13318-background	RA: 14 19 10.8881 (214.7953671d) Dec: +52 47 19.85 (52.78885d) Equinox: J2000										
	Comments: Category=Calibration Description=[Target acquisition test]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
	F560W	All MRS			YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1130W	FASTR1	50	4	1	Dither 1	2	8	1126.666	
	1	LONG(C)	MRSLONG		SLOWR1	25	1	1	Dither 1	2	2	1194.496	
	1	LONG(C)	MRSSHORT		SLOWR1	25	1	1	Dither 1	2	2	1194.496	

Special Requirements	Group Observations 3, 5, Non-interruptible
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